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1 Einführung

TODO Englisch, Problem umstellen, Interpretation und Motivation
Wir betrachten das dynamische Transportproblem

$$\inf \int_0^1 \int_\Omega \frac{|J_t|^2}{2\rho_t} dxdt + \int_0^1 \int_\Omega \frac{|f_t|^2}{2\rho_t} dxdt + \int_0^1 \int_\Gamma |\rho(t) - \rho_d|^2 dxdt$$

mit Kurve $\Gamma \subset (\Omega \times (0,1))$ und Nebenbedingungen

$$\frac{\partial \rho_t}{\partial t} + \nabla \cdot (J_t + \rho_t v_t) = f_t \quad auf \Omega \times (0,1)$$
$$(J_t + \rho_t v_t) \cdot n = 0 \quad \partial\Omega \times (0,1).$$

Dabei bezeichnet J_t das Energiefunktional zum Zeitpunkt t, f_t ist die Quelle, ρ_t ist ???, v_t das Vektorfeld und Datenterm. Mit Neumann-Randbedingungen haben wir halt ein abgeschlossenes System.
Um dieses Problem zu lösen, benötigt es einige Grundlagen aus Maßtheorie und Funktionalanalysis, eine Einführung in optimalen Transport und PDE/Variationsrechnung/Konvexe Analysis. Dann erarbeiten wir Existenzresultate und lösen das Problem noch numerisch.
Gegeben ein Vektorfeld und einer Datenkurve, welche auch eine Dichte im gewissen Sinne ist: Was war der Ursprung?
Bisherige Quellen: Santambrogio 4.1.2 (und OT-Theorie), Folland Maßtheorie + Funkana, Alt Funkana, OT von Figalli, Gradient Flows von Ambrosio/Gigli, Evans PDE, Benamou Brenier Paper, Variationsrechnung Skript Schmidt und Struwe

2 Mathematical Background

TODO Überlegen: Was brauche ich auch Maßtheorie, Konvexer Analysis, Funkana, Variationsrechnung... Wenn es eher ein Vorgehen ist wie Testfunktionen und PI oder KKT dann in eine Box packen. Das auch alles auf Englisch.
Das zu behandelnde Thema benötigt Kenntnisse und Grundlagen aus unterschiedlichsten mathematischen Teilgebieten. So geben wir einen Überblick über Maßtheorie, insbesondere Radonmaße und den Satz von Radon-Nikodym, grundlegende Definitionen aus der Funktionalanalysis - unterschiedliche Konvergenzarten (schwach und schwach-*) und in diesem Zug auch Dualräume. Darüber hinaus brauchen wir noch grundlegende Resultate aus der Funktionalanalysis und PDE-Theorie, etwa Lösungstheorie und Methoden der Variationsrechnung. Hier auch noch Konvexe Analysis?

2.1 Maßräume

2.1.1 Grundlegende Definitionen und Eigenschaften

Im Kontext von Transporttheorie, wo uns interessiert, wie wir möglichst effizient Masse von einem zum anderen Standort transportieren können, benötigen wir eine Idee von dem Konzept Masse und wie diese quantifiziert werden kann. Naheliegend finden wir dazu Antworten in der Maßtheorie. Wir beschäftigen uns erst mit dem Begriff eines Maßes und spezielle Klassen dieses Begriffs sowie zentrale Eigenschaften von Elementen, die wir messen können und überlegen uns, in welchem Sinne eine Ableitung für Maße existiert.

Definition 1 (σ-Algebra)**.** Sei X eine Menge. Ein Mengensystem $\mathcal{A}$ über X heißt **σ-Algebra**, wenn die Eigenschaften

1. $\emptyset \in \mathcal{A}$,
2. $A \in \mathcal{A} \implies A^C \in \mathcal{A}$,
3. $(A_i)_{i\in I} \in \mathcal{A} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$,

erfüllt sind.

Definition 2 (Signiertes Maß)**.** Eine Abbildung $\mu : \mathcal{A} \to (-\infty, \infty]$ heißt **signiertes Maß**, wenn gilt

1. $\mu(\emptyset) = 0$,
2. $\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$

für paarweise disjunkte A_i. Zusätzlich nennen wir Abbildungen $\tilde{\mu} : \mathcal{A} \to [0, \infty]$ mit den gleichen Eigenschaften *positive Maße* und Abbildungen $\mathbb{P} : \mathcal{A} \to [0, 1]$ heißen *Wahrscheinlichkeitsmaße*.

Den Raum $(X, \mathcal{A})$ bezeichnen wir als **Messraum** und Elemente des Maßraums heißen *messbare Mengen*. Messbare Mengen E mit der Eigenschaft $\mu(E) = 0$ heißen *Nullmengen* bezüglich μ oder auch $\mu-$Nullmengen. Ist es möglich, den Raum X als abzählbare Vereinigung von Mengen $A_1, \ldots, A_n$ mit $\mu(A_i) < \infty$ für alle $i \in \mathbb{N}$ zu schreiben, so heißt das Maß *σ-endlich*. Ein Messraum ausgestattet mit einem Maß, also $(X, \mathcal{A}, \mu)$ heißt **Maßraum**.

Definition 3 (Erzeugte σ-Algebra)**.** Sei $\mathcal{M} \subset \mathcal{P}(X)$. $\sigma(\mathcal{M}) = \bigcap_{\mathcal{A}\in\mathcal{F}(\mathcal{M})} \mathcal{A}$ heißt die *erzeugte σ-Algebra*, wobei $\mathcal{F}(\mathcal{M}) := \{\mathcal{M} \subseteq \mathcal{A} \mid \mathcal{A} \text{ ist } \sigma\text{-Algebra auf } X\}$.

Definition 4 (Borel-σ-Algebra)**.** Sei $(X, \mathcal{O})$ ein topologischer Raum, wobei $\mathcal{O}$ das Mengensystem der offenen Mengen ist. $\sigma(\mathcal{O})$ ist die *Borelsche-σ-Algebra* und wir schreiben $\mathcal{B}(X) := \sigma(\mathcal{O})$.

Eine Klasse von besonders netten Maßen sind sogenannte Radon-Maße. Diese sind intuitiv, da kompakten Mengen ein endliches Maß zuordnen und sich Mengen auch approximieren lassen.

Definition 5 (Radon-Maße)**.** Ein signiertes Maß $\mu : \mathcal{A} \to [-\infty, \infty)$ heißt Radon-Maß, wenn gilt

- $\mu(K) < \infty$ für $K \in \mathcal{A}$ kompakt
- Für jedes $E \in \mathcal{A}$ existiert ein $B \in \mathcal{B}(X)$ mit $B \supseteq E$ und $\mu(B) = \mu(E)$

TODO sigma-additiv, $X = \mathbb{R}^d$ definieren oder allgemeiner - wollen wir Polish Spaces oder brauchen wir bestimmte Eigenschaften?

Definition 6 (Variationsmaß)**.** Für ein σ-additives signiertes Maß $\mu : \mathcal{A} \to [-\infty, \infty)$ definieren wir das Totalvariationsmaß

$$|\mu|(B) := \sup\left\{\sum_{i=1}^{\infty} |\mu(B_i)| \,\middle|\, \dot{\bigcup}_{i=1}^{\infty} B_i = B\right\} \quad B \in \mathcal{B}(X).$$

Mit $\|\mu\|_{TV} = |\mu|(X)$ definieren wir die Totalvariations-Norm.

Prop? Lemma? Eig? Es gilt für alle $B \in \mathcal{B}(X)$

$$|\mu|(B) = \inf\{|\mu|(U) \mid B \subset U, U \text{ offen}\}$$
$$|\mu|(B) = \sup\{|\mu|(K) \mid B \supset K, K \text{ kompakt}\}$$

TODO: Zeige, dass durch diese Norm ein Banachraum definiert ist – an späterer Stelle dann Dualraum zeigen. Hier auch noch vektorwertige Maße?
Für maßwertige Lösungen von PDE benötigen wir den zentralen Satz von Radon-Nikodym. Hierzu braucht es vorbereitende Definitionen

Definition 7 (Absolutstetigkeit von Maßen)**.** Ein Maß μ heißt absolutstetig bezüglich ν mit Notation $\mu \ll \nu$, wenn für alle $A \subset \Omega$ mit $\nu(A) = 0$ auch $\mu(A) = 0$ gilt.

Zusätzlich haben wir mehr oder weniger das Gegenteil von Absolutstetigkeit.
TODO andere Definition verwenden?

Definition 8 (Singularität von Maßen)**.** Seien μ, ν signierte Maße auf $(X, \mathcal{A})$. Wir nennen die zwei Maße singulär, wenn Mengen $E, F \in \mathcal{A}$ existieren mit $E \cap F = \emptyset, E \cup F = X$ und $\mu(E) = 0, \nu(F) = 0$. Wir schreiben dabei $\mu \perp \nu$.

Theorem 2.1.1 (Hahn-Zerlegung)**.** *Sei μ ein signiertes Maß auf $(X, \mathcal{A})$, dann existieren Mengen P mit $\mu(P) \geq 0$ und N mit $\mu(N) \leq 0$ für die $P \cup N = X$ und $P \cap N = \emptyset$ gilt.*

Beweis S 99

Theorem 2.1.2 (Jordan-Zerlegung)**.** *Sei μ ein signiertes Maß, dann existieren zwei positive Maße μ^+, μ^-, so dass $\mu = \mu^+ - \mu^-$ und $\mu^+ \perp \mu^-$.*

Hier noch Beweis

Theorem 2.1.3 (Radon-Nikodym)**.** *Sei ν ein signiertes Maß und μ ein positives Maß, die jeweils σ-endlich auf $(X, \mathcal{A})$ sind. Es existieren eindeutige σ-endliche signierte Maße λ, ρ auf $(X, \mathcal{A})$ mit*

$$\lambda \perp \mu,$$
$$\rho \ll \mu,$$
$$\nu = \lambda + \rho.$$

Zusätzlich existiert eine μ-integrierbare Funktion $f : X \to \mathbb{R}$ mit $d\rho = f d\mu$. f hießt dabei Radon-Nikodym-Ableitung und wird geschrieben als $\frac{d\nu}{d\mu}$ // $d\nu = \frac{d\nu}{d\mu} d\mu$.
Es gilt $g \in L^n(\nu) \implies g(d\nu/d\mu) \in L^1(\mu)$ und $\int g d\nu = \int g \frac{d\nu}{d\mu} d\mu$.

Der Raum von Maßen ist konvex (wieso? Das ist super wichtig zu verstehen) Wegen Radon-Nikodym lässt sich jedes absolutstetige Maß schreiben als $\rho = f(x)dx$ mit $f \in L^1, \int f = 1$ (WMaß) und $L^1 \subset \mathcal{P}$ TODO. Vektor-wertige Maße?

2.2 Functional Analysis TODO

Nachdem das Problem das Konzept von Operatoren auf unendlichdimensionalen Räumen benötigt, führen wir zunächst grundlegende Begriffe wie Konzepte ein. Im Folgenden ist X dabei ein normierter $\mathbb{K}$-Vektorraum. Ist X vollständig, so heißt er *Banachraum*. Enthält X eine abzählbare Teilmenge A mit $\overline{A} = X$, dann nennen wir X *separabel*. Ist zusätzlich auch Y ein normierter Vektorraum, so bezeichnen wir mit $L(X;Y)$ die Menge der stetigen linearen Abbildungen $T : X \to Y$, wobei wir T einen Operator nennen. Ein grundlegendes Resultat der Funktionalanalysis ist, dass lineare Operatoren genau dann stetig sind, wenn sie beschränkt sind. Mit X' bezeichnen wir den *Dualraum* von X. Elemente von X' sind dabei lineare Funktionale $\varphi : X \to \mathbb{K}$ (als Beispiel Integrale nennen). Dieses Objekt erlaubt es uns, zwei weitere Konvergenzarten zu definieren.
Wir nennen eine Folge $(x_n)_{n\in\mathbb{N}} \subset X$ *stark konvergent*, falls $\|x_n\|_X \overset{n\to\infty}{\longrightarrow} \|x\|_X$.
Wir nennen eine Folge $(x_n)_{n\in\mathbb{N}} \subset X$ *schwach konvergenz*, falls für alle $\varphi \in X'$ gilt $\varphi(x_n) \overset{n\to\infty}{\longrightarrow} \varphi(x)$.
Wir nennen eine Folge $(\varphi_n)_{n\in\mathbb{N}} \subset X'$ *schwach-* konvergent*, falls für alle $x \in X$ gilt $\varphi_n(x) \overset{n\to\infty}{\longrightarrow} \varphi(x)$.
Hier interessiert uns die Konvergenz von (Wahrscheinlichkeits-)Maßen(TODO Motivation).

Theorem 2.2.1. *Banach-ALagolu*

Theorem 2.2.2. *Riesz Representation*

Daraus dann folgern dass $C_0'(\Omega) = \mathcal{M}(\Omega)$ "Maße sind Lineare Funktionale im gewissen Sinne", braucht es für Duale Formulierung und Variationsprobleme
Reflexivität, Hahn-Banach,
Konvergenzarten von W-Maßen/Dichten im OT Kontext: narrow convergence, tightness, prokhorov, portmanteau? Generell: Was darf ich voraussetzen, was muss ich beweisen und was kann ich nur einmal erwähnen?

2.3 Variationsrechnung

Das vermutlich Verschieben nach 4: Warum brauchen wir das (Motivation), was sind die Methoden, die wir benötigen? Resultate verwenden für Lösung von Dualer Formulierung OT und DOT
PDE-Theorie und damit Lp? Lp Def, Hölder, Minkowski, Dualität; Sobolew, schwache Ableitung, Einbettung, Poincare; Distributionen, Testfunktionen, schwache Formulierung von PDE, Lax-Milgram
Variationsrechnung: Korezivität, Kompaktheit, lsc, Euler-Lagrange Gleichung und Lagrange Multiplikatoren (Direkte Methode

3 Dynamical Optimal Transport

In this chapter we lay out the framework in which our mathematical model is presented. We introduce the concept of optimal transport, first the static formulation of Monge-Kantorovich and in combination with the continuity equation the dynamical formulation that is then presented through the lense of the Benamou-Brenier formulation. The goal then is to establish a result on the equivalence of solutions and also introduce a version of the ALG2 algorithm presented in (BB). This helps us understand the approach used to solve the modified variational problem in the next chapter.

3.1 Optimal Transport Theorie

The central question of the problem was proposed by Gaspard Monge in 1781. Given a pile of dirt and an equal sized hole, how can it be filled "optimally" (Tikz Grafik einfügen). This raises questions such as what optimal means in that context and also if the problem can even be solved.

Definition 9 (Monge-Problem)**.** Let $\mu \in \mathscr{P}(X)$ and $\nu \in \mathscr{P}(Y)$ be two probability measures and $c : X \times Y \to [0, \infty]$ a continuous or lower semi-continuous *cost function*. The **Monge problem** is to find a measurable map $T : X \to Y$ that minimizes the total transpotation cost

$$\inf \left\{ \int_X c(x, T(x)) \, \mathrm{d}\mu(x) \;\middle|\; T_{\#}\mu = \nu \right\}. \tag{MP}$$

This initial definition allows for an intuitive interpration of the problem, however it inherently comes with the problem that it may not exist a solution.

Example 1. Consider the case $X = Y = \mathbb{R}^n$ and $c(x, y) = \|x - y\|^2$, where $\mu, \nu \in (\mathbb{R}^n)$ have smooth densities with respect to the Lebesgue measure. If μ is not absolutely continuous (e.g. if it has an atom), a transport map T solving (**??**) may fail to exist: mass concentrated at a single point cannot be split and sent to multiple locations to match ν. One simply consideres the case of Dirac measures.

Include tikz mit Dirac Mass
Another issue may be that even if a transport map exists, it may not be unique, if we do not allow for strict regulaztion. These potential problems motivate a *relaxation*, that is, rather than seeking a *transport map* we allow *tansport plans*, which are measures on the product space $X \times Y$ that have correct marginals. This idea was introduced by Leonid Kantorovich about 150 years later and is foundational to the modern theory of optimal transport. First we define the concept of a coupling, which formalizes the restriction on the marginals.

Definition 10 (Coupling)**.** Let $\mu \in \mathscr{P}(X)$ and $\mu \in \mathscr{P}(Y)$ be probability measures. A measure $\gamma \in \mathscr{P}(X \times Y)$ is called a **coupling** of μ and ν if its marginals are μ and ν, respectively. That is,

$$(\pi_X)_{\#}\gamma = \mu \quad \text{and} \quad (\pi_Y)_{\#}\gamma = \nu,$$

where $\pi_X(x, y) = x$ and $\pi_Y(x, y) = y$ denote the canonical projections onto X and Y. The set of all such couplings is denoted by $\Gamma(\mu, \nu)$. Couplings are commonly referred to as **transport plans**.

A result that is crucial for upcoming result is the fact, that under suitable conditions, the set Γ is non-empty.

Lemma 3.1.1. *Let X be a Polish space and $T : X \to Y$ a measurable map such that $T_{\#}\mu = \nu$. Then $\Gamma(\mu, \nu)$ is non-empty, convex and tight. Moreover, $\Gamma(\mu, \nu) \subset \mathscr{P}(X \times Y)$ is compace under the topology of narrow convergence.*

Proof. Prokhorok, marginals of tight families are tight. □

Definition 11 (Kantorovich Problem)**.** Given a lower semi-continuous cost function $c : X \times Y \to [0, \infty]$ and probability measures $\mu \in \mathscr{P}(X)$, $\nu \in \mathscr{P}(Y)$, the **Kantorovich problem** seeks to minimize the total cost over all couplings

$$\inf \left\{ \int_{X \times Y} c(x, y) \, \mathrm{d}\gamma(x, y) \;\middle|\; \gamma \in \Gamma(\mu, \nu) \right\}.$$

Remark 1. Since for any feasible transport map T we have

$$\int_X c(x, T(x)) \, \mathrm{d}\mu(x) = \int_X c \circ (\mathrm{Id} \times T)(x) \, \mathrm{d}\mu(x) = \int_{X \times Y} c(x, y) \, \mathrm{d}\gamma_T(x, y),$$

which implies that the Monge problem is a special case of the Kantorovich problem. As a consequence it holds

$$C_M(\mu, \nu) \geq C_K(\mu, \nu),$$

where C_M and C_K denote the optimal costs of the Monge and Kantorovich problems, respectively.

Tikz zu Kantorovich einfügen
Under mild assumptions we can proof the existence of a solution for the static optimal transport problem.

Theorem 3.1.2 (Existence of Optimal Transport Plans)**.** *Let $c : X \times Y \to [0, \infty]$ be lower semi-continuous and let $\mu \in \mathscr{P}(X)$, $\nu \in \mathscr{P}(Y)$ be probability measures on Polish spaces. Then there exists a coupling $\overline{\gamma} \in \Gamma(\mu, \nu)$ that solve the Kantorovich problem, i.e.*

$$\int_{X \times Y} c(x, y) \, \mathrm{d}\overline{\gamma}(x, y) = \inf_{\gamma \in \Gamma(\mu, \nu)} \int_{X \times Y} c(x, y) \, \mathrm{d}\gamma(x, y).$$

TODO Box für direkte Methode einfügen

Proof. Assume that

$$\alpha = \inf_{\gamma \in \Gamma(\mu, \nu)} \int_{X \times Y} c(x, y) \, \mathrm{d}\gamma(x, y) < +\infty,$$

otherwise, every $\gamma \in \Gamma(\mu, \nu)$ is trivially a minimizer. Let $(\gamma_k)_{k \in \mathbb{N}} \subset \Gamma(\mu, \nu)$ be a sequence such that

$$\int_{X \times Y} c(x, y) \, \mathrm{d}\gamma_k(x, y) \overset{k \to \infty}{\longrightarrow} \alpha.$$

Since $\Gamma(\mu, \nu)$ is a tight set, the sequence $(\gamma_k)_{k \in \mathbb{N}} \subset \Gamma(\mu, \nu)$ is tight. By Prokhorovs theorem, there exists a subsequence $(\gamma_{k_j})_{j \in \mathbb{N}}$ that converges narrowly to some measure $\overline{\gamma}$ (Hier Resultat staten). By semi-continuity of the cost functional we have

$$\alpha = \inf_{\gamma \in \Gamma(\mu, \nu)} \int_{X \times Y} c(x, y) \, \mathrm{d}\gamma(x, y) = \liminf_{j \to \infty} \int_{X \times Y} c(x, y) \, \mathrm{d}\gamma_{k_j}(x, y) \geq \int_{X \times Y} c(x, y) \, \mathrm{d}\overline{\gamma}(x, y).$$

But since $\overline{\gamma} \in \Gamma(\mu, \nu)$, we also have

$$\int_{X \times Y} c(x, y) \, \mathrm{d}\overline{\gamma} \geq \alpha,$$

which implies that $\overline{\gamma}$ is a minimizer.

□

Hier noch Santambrogio 1.17.

3.2 Dynamical Optimal Transport

We want to introduce the concept of dynamical optimal transport. In contrast to static optimal transport, we do not only care about finding an optimal map or optimal coupling to transport one distribution to another but are interested in a sense of flow in time, where for each time step $t \in (0, T)$ the flow along a vector field is optimal. The necessary tools for that are a definition of a new metric, namely the Wasserstein metric, a new sense of derivative, namely a metric derivative and a little bit of convex analysis. We show that solutions of dynamic and static optimal transport are in fact equivalent and present a numerical method for solving the dynamical optimal transport problem, including an implementation.
While the total variation measure indeed takes into account the difference between two probability measures, it ignores a spacial distance, making it unfit for our purposes. A better choice is the Wasserstein-Distance.

Definition 12 (Wasserstein Distance)**.** Let X be a separable metric space, where each element $\mathscr{P}(X)$ is probability and hence a Radon measure and let $p \geq 1$. We define the p-th Wasserstein distance between two probability measures $\mu, \nu \in \mathscr{P}(X)$ as

$$W_p^p(\mu, \nu) := \inf \left\{ \int_{X^2} d(x, y)^p \, \mathrm{d}\gamma(x, y) \;\middle|\; \gamma \in \Gamma(\mu, \nu) \right\}.$$

TODO W ist eine Metrik
The concept of dynamic optimal transport is linked to the continuity equation, that can be interpreted as a condition to preserve mass during the flow of particles across a given velocity field and also allows us to then follow each single time step. Introducing the variable $\rho(t, x)$ with $t \in (0, 1)$ and $x \in \Omega$ is essentialy the flow of our density in time-space with $\rho(0, x) = \mu(x)$ and $\rho(1, x) = \nu(x)$.
TODO schöner formulieren
The direction of a possible flow at time t and in x is given by a velocity field $v(t, x)$. We introduce lagrangian coordinates and denote by $X(t, x)$, $x \mapsto x_t$ the map of the position of some particle x to its position at time t, which can easily be applied to sets as well. The flow of each particle can be obtained by following the particle along the velocity field and we write

$$\frac{\partial X}{\partial t}(t, x) = v(t, X(t, x)), \quad X(0, x) = x$$

and when using sets specifically

$$\int_E \mu(x) \, \mathrm{d}x = \int_{X(t,E)} \rho(t, x) \, \mathrm{d}x \quad \forall t \in (0, 1).$$

The above formulation leads us to the realization that by change of variables we have

$$X(t,\cdot)_{\#}\mu = \rho(t,\cdot),\ X(1,\cdot)_{\#}\mu = \nu.$$

Our goal is to rewrite the Wasserstein distance such that it merely depends on ρ, v and has the form

$$W_2^2(\mu,\nu) = \inf_{\rho,v} \int_0^1 \int_\Omega \rho(t,x)|v(t,x)|^2 \mathrm{d}x\mathrm{d}t$$
$$\text{s.t.} \begin{cases} \partial_t \rho_t + \nabla \cdot (\rho v) = 0, \\ \rho(0,x) = \mu, \\ \rho(1,x) = \nu. \end{cases}$$

The first constraint is known as he continuity equation that can be interpreted as mass conservation. We begin to show the equivalence by calculating

$$\begin{aligned} W_2^2(\mu,\nu) &\le \int_\Omega \mu(x)|X(1,x) - X(0,x)|^2 \,\mathrm{d}x \\ &\overset{\text{FTC}}{=} \int_\Omega \mu(x) \left| \int_0^1 \frac{\partial X}{\partial t}(t,x)\mathrm{d}t \right|^2 \,\mathrm{d}x \\ &\overset{\Delta}{\le} \int_\Omega \int_0^1 \mu(x) \left| \frac{\partial X}{\partial t}(t,x) \right|^2 \mathrm{d}t\mathrm{d}x \\ &= \int_0^1 \int_\Omega \mu(x)|v(t,X(t,x))|^2 \,\mathrm{d}x\mathrm{d}t. \end{aligned}$$

To evaluate the inner integral more easily, we define

$$S(x) = X(t,x), \quad S_{\#}\mu = \rho(t,x), \quad \varphi(y) = |v(t,y)|^2$$

and using change of variables yields

$$\begin{aligned} \int_\Omega \mu(x)|v(t,X(t,x))|^2 \,\mathrm{d}x &= \int_\Omega \mu(x)\varphi(S(x)) \,\mathrm{d}x \\ &= \int_\Omega \rho(t,y)\varphi(y) \,\mathrm{d}y \\ &= \int_\Omega \rho(t,y)|v(t,y)|^2 \,\mathrm{d}y \end{aligned}$$

and hence we have

$$W_2^2(\mu,\nu) \le \int_0^1 \int_\Omega \rho(t,x)|v(t,x)|^2 \,\mathrm{d}x\mathrm{d}t$$

for all ρ, v that satisfy

$$\begin{cases} \partial_t \rho_t + \nabla \cdot (\rho v) = 0, \\ \rho(0,x) = \mu, \\ \rho(1,x) = \nu. \end{cases}$$

Equality is obtained, if $X(1,x)$ is an optimal transport from μ to ν and $\frac{\partial X}{\partial t} = u(x)$ is constant in time. Assuming constant flow in time we simplify

$$X(t,x) = x + u(x)t \implies X(1,x) = x + u(x) =: T(x)$$

and solving for u gives us

$$\begin{aligned} u(x) = T(x) - x &\implies X(t,x) = x + (T(x) - x)t = x(1-t) + tT(x) \\ &\implies v(t, X(t,x)) = u(x) = (T - I)(x). \end{aligned}$$

Therefore we found a formulation for an optimal transportation by

$$v(t,y) = (T - I)[(1-t)I + tT]^{-1}(y)$$

and validated the problem formulation above. We can therefore write the optimization problem as

$$W_2^2(\mu,\nu) = \inf_{\rho,v} \int_0^1 \int_\Omega \rho(t,x)|v(t,x)|^2 \,\mathrm{d}x\mathrm{d}t$$
$$s.t. \begin{cases} \partial_t \rho_t + \nabla \cdot (\rho v) = 0, \\ \rho(0,x) = \mu, \\ \rho(1,x) = \nu. \end{cases}$$

TODO Das untere brauchen wir hier nicht mehr?
We try to solve the Lagrangian saddle-point problem. Note that by multiplying test functions $\varphi \in C_c^\infty((0,1) \times \Omega)$ and partial integration, we obtain the weak form of the continuity equation

$$\int_\Omega (\nu\varphi(1,x) - \mu\varphi(0,x)) \,\mathrm{d}x\mathrm{d}t - \int_0^1 \int_\Omega (\rho\varphi_t + \rho v \cdot \nabla\varphi) \,\mathrm{d}x\mathrm{d}t$$

and hence the Lagrangian has the form

$$\mathcal{L}(\varphi,\rho,v) = \int_0^1 \int_\Omega \left(\frac{\rho|v|^2}{2} - \rho\varphi_t - \rho v \cdot \nabla\varphi \right) \mathrm{d}x\mathrm{d}t + \underbrace{\int_\Omega (\nu\varphi(1,x) - \mu\varphi(0,x)) \,\mathrm{d}x}_{G(\varphi)}.$$

By defining $m := \rho v$ we rewrite

$$\mathcal{L}(\varphi,\rho,m) = \int_0^1 \int_\Omega \left(\frac{|m|^2}{2\rho} - \rho\varphi_t - m \cdot \nabla\varphi \right) \mathrm{d}x\mathrm{d}t + G(\varphi).$$

3.3 Equivalence of the Static and Dynamic Formulation

The goal here is to show the relationship between static optimal transport, that is the Monge-Kantorovich formulation, and dynamical optimal transport. We will find that having a solution for one of these problems will then give us a solution to the other, making the problems equivalent. For this purpose, we define a few key terms.

Definition 13 (Metric Derivative)**.** Let (X, d) be a metric space. We define the derivative of a curve $\gamma : [0, 1] \to X$ as

$$|\gamma'|(t) := \lim_{h \to 0} \frac{d(\gamma(t+h) - \gamma(t))}{|h|}. \tag{1}$$

Note that the absolute value of the denominator is necessary for the expression to be well-defined.

Definition 14 (Absolutely Continuous Curves)**.** A curve $\gamma : [0, 1] \to X$ is said to be absolutely continuous, if there exists a $g \in L^1([0, 1])$ such that the inequality

$$d(\gamma(t), \gamma(s)) \leq \int_t^s g(x)dx \tag{2}$$

holds for all $t < s$. If this property holds, we write $\gamma \in \mathrm{AC}([0, 1], X)$.

To show the first main result of this chapter, we need some auxiliary results.

Lemma 3.3.1. *Let $\gamma \in \mathrm{AC}([0, 1], X)$ and $|\gamma'|$ be its metric derivative. For $t < s$ we get*

$$d(\gamma(t), \gamma(s)) \leq \int_t^s |\gamma'|(x)\mathrm{d}x.$$

Proof. Let $\gamma \in \mathrm{AC}([0, 1], X)$. Fix $t < s$ and define

$$f(r) := d(\gamma(t), \gamma(r)), \quad r \in [t, s].$$

Notice that by absolute continuity of γ and Lipschitz continuity of $x \mapsto d(\gamma(t), x)$ the map f is also absolutely continuous. By the reverse triangle equality we obtain

$$|f(r+h) - f(r)| = |d(\gamma(r+h), \gamma(t)) - d(\gamma(t), \gamma(r))| \leq d(\gamma(r+h), \gamma(r))$$

and therefore

$$\lim_{|h| \to 0} \left| \frac{f(r+h) - f(r)}{h} \right| \leq \lim_{|h| \to 0} \frac{d(\gamma(r+h), \gamma(r))}{|h|} \implies |f'(r)| \leq |\gamma'|(r)$$

which holds for almost every $r \in [t, s]$. By absolute continuity and the fundamental theorem of calculus we get

$$\int_t^s f'(r)\,\mathrm{d}r = f(s) - f(t) = d(\gamma(t), \gamma(s))$$

and it follows

$$d(\gamma(t), \gamma(s)) = \int_t^s f'(r)\,\mathrm{d}r \leq \int_t^s |f'(r)|\,\mathrm{d}r \leq \int_t^s |\gamma'(r)|\,\mathrm{d}r.$$

□

BOX: By the Rademacher-Theorem we conclude that the metric dericative may exist for a.e. $t \in [0, 1]$ if the curve is Lipschitz-continuous. The next results guarantees us that it is possible to reparametrize absolutely continuous curves to be Lipschitz-curves.

Lemma 3.3.2. *If $\gamma \in \mathrm{AC}([0, 1], X)$, then γ admits a Lipschitz reparametrization.*

Proof. Let $\gamma : [0,1] \to X$ be absolutely continuous and define its accumulated length by

$$\ell(t) = \int_0^t |\gamma'|(r)\,\mathrm{d}r \quad t \in [0,1].$$

Then ℓ is continuous, nondecreasing and

$$\ell(0) = 0, \quad \ell(1) = L(\gamma).$$

If $L(\gamma) = 0$, then γ is constant, hence already Lipschitz. Therefore we may assume $L(\gamma) > 0$. For $\sigma \in [0, L(\gamma)]$, define

$$\tau(\sigma) := \min\{t \in [0,1] : \ \ell(t) = \sigma\}.$$

By continuity of ℓ this is well-defined. Now define the reparametrized curve

$$\tilde{\gamma} : [0, L(\gamma)] \to X, \quad \tilde{\gamma}(\sigma) := \gamma(\tau(\sigma)).$$

Let $0 \le \sigma_1 \le \sigma_2 \le L(\gamma)$. Since ℓ is non-decreasing and γ is absolutely continuous, we obtain

$$\begin{aligned} d(\tilde{\gamma}(\sigma_1), \tilde{\gamma}(\sigma_2)) &= d(\gamma(\tau(\sigma_1)), \gamma(\tau(\sigma_2))) \\ &\le \int_{\tau(\sigma_1)}^{\tau(\sigma_2)} |\gamma'|(r)\,\mathrm{d}r \\ &= \ell(\tau(\sigma_2)) - \ell(\tau(\sigma_1)) \\ &= \sigma_2 - \sigma_1. \end{aligned}$$

□

Lemma 3.3.3. *For $p > 1$ the optimal Transport map between two absolutely continuous measures exists. TODO?*

Proof. Beweis in 3.1 mit Kantorovich Dualität / Subdifferential – bereits im Seminar gemacht □

TODO hier noch Lemmata für das Theorem: Weak und Distributional ist gleich (Prop 4.2 Sant bzw ganzes Kapitel 4.1), Lemma 5.2
One of the main theorems of this chapter is the following.

Theorem 3.3.4. *Let $(\gamma_t)_{t\in[0,1]} \subset \mathrm{AC}(\mathscr{P}_p(X))$ be a family of absolutely continuous curves on the space of probability measures. For a.e. $t \in [0,1]$ there exists a velocity field $v_t \in L^p(c_t, \mathbb{R}^d)$ with the properties*

- *the continuity equation is satisfied in the distributional sense*
- *we have $\|v_t\|_{L^p} \le |\gamma'|(t)$ for a.e. $t \in [0,1]$.*

The idea of the proof is to construct a sequence of γ_t and a corresponding sequence for v_t such that the continuity equation holds, γ_t^k is absolutely continuous for all k and the limit admits the desired inequality.

Proof. By Lemma **??** we assume μ to be Lipschitz. We define the sequence $\mu_t^k := \mu_t * \eta_k$, where η_k is a Dirac sequence supported on $B\left(0, \frac{1}{k}\right)$. This is a necessary step, since for example $\delta_x \in \mathscr{P}_2(X)$ is not absolutely continuous. Since

$$\operatorname{supp}(\mu_t * \eta_k) \subset \Omega + B\left(0, \frac{1}{k}\right)$$

we extend the domain to a convex set $\Omega' \supset \Omega$. According to Lemma **??**, for $t = \frac{i}{k}$, $i = 0, \dots, k$ the optimal transport map $T^{i,k} : \Omega' \to \Omega'$ from $\mu^k_{\frac{i}{k}}$ to $\mu^k_{\frac{i+1}{k}}$ exists as a consequence of absolute continuity. Defining the linear interpolation with constant speed

$$\mu_t^k := \left((i+1-tk)\mathrm{id} + (kt-i)T^{i,k}\right)_{\#} \mu^k_{\frac{i}{k}}$$

and deriving with respect to t yields the velocity

$$v^{i,k} := k(T^{i,k}(x) - x).$$

Since the continuity equation expects us not to have a particle indexed by its starting position but by the current position, we define

$$T_t^{i,k} := (i+1-kt)\mathrm{id} + (kt-i)T_k^{ik},$$

giving us the desired velocity field

$$v_t^k := v_{i,k}((T_t^{i,k})^{-1}).$$

For the interval $\left(\frac{i}{k}, \frac{i+1}{k}\right)$, define

$$X_t(x) := T_t^{i,k}, \quad \mu_t^k = (X_t)_{\#}\mu^k_{\frac{i}{k}}$$

and by definition we obtain the ordinary differential equation

$$v_t^k(X_t(x)) = v_{i,k} = X_t'(x).$$

Taking test functions and integrating by parts gives

$$\begin{aligned}
&\int \varphi(y)\,\mathrm{d}\mu_t^k(y) = \int \varphi(X_t(x))\,\mathrm{d}\mu^k_{\frac{i}{k}}(x) \\
\iff &\frac{d}{dt}\int \varphi(y)\,\mathrm{d}\mu_t^k(y) = \int \nabla\varphi(X_t(x)) \cdot v_t^k(X_t(x))\,\mathrm{d}\mu^k_{\frac{i}{k}}(x) \\
\iff &\frac{d}{dt}\int \varphi\,\mathrm{d}\mu_t^k = \int \nabla\varphi(y) \cdot v_t^k(y)\,\mathrm{d}\mu_t^k(y),
\end{aligned}$$

which is the weak form of the continuity equation, hence the pair (μ_t^k, v_t^k) satisfies

$$\partial_t \mu_t^k + \nabla \cdot (v_t^k \mu_t^k) = 0$$

in the distributional sense.

To estimate an upper-bound for the newly obtained veloctiy field in order to extract a weak limit, we compute

$$\begin{aligned}\|v_t^k\|_{L^p(\mu_t^k)}^p &= \int_\Omega |v^{i,k}|^p \,\mathrm{d}\mu_{\frac{i}{k}}^k = k^p W_p^p\left(\mu_{\frac{i}{k}}^k, \mu_{\frac{i+1}{k}}^k\right)\\ &\le k^p W_p^p\left(\mu_{\frac{i}{k}}, \mu_{\frac{i+1}{k}}\right)\\ &\le k^p\left(\int_{\frac{i}{k}}^{\frac{i+1}{k}} |\mu'|(s)\,\mathrm{d}s\right)^p\\ &\overset{(*)}{\le} k\int_{\frac{i}{k}}^{\frac{i+1}{k}} |\mu'|^p(s)\,\mathrm{d}s \le \mathrm{Lip}(\mu)\end{aligned}$$

$(*)$ is obtained by the Jensen inequality where the last inequality by the assumtption that the curve is Lipschitz. Also we can estimate

$$\int_a^b \|v_t^k\|_{L^p(\mu_t^k)}^p \,\mathrm{d}t \le \int_a^b |\mu'|^p(s)\,\mathrm{d}s + \frac{2\mathrm{Lip}(\mu)}{k}.$$

This can be seen by taking an equidistant partition $\left\{\frac{i_a}{k},\dots,\frac{i_b}{k}\right\} \subset [a,b]$ and splitting the integral into $\int_a^{i_a} + \int_{i_a}^{i_b} + \int_{i_b}^b$. The middle integral uses the estimate $(*)$, the other two use the Lipschitz-bound provided in the previous calculation. Adding the fact, that $i_a - a$, $b - i_b \le \frac{1}{k}$ gives the desired result. We conclude that $v_t^k \in L^p(\mu_t^k)$.
Define now vector measures E^k with test velocity field $\phi : \Omega' \times [0,1] \to \mathbb{R}^d$ by

$$\int \phi(t,x)\cdot \,\mathrm{d}E^k(t,x) := \int_0^1 \int_{\Omega'} \phi(t,x)\cdot v_t^k(x)\,\mathrm{d}\mu_t^k(x)\mathrm{d}t$$

and compute the total variation

$$\|E\| = \int_0^1 \|v_t^k\|_{L^1(\mu_t^k)}\,\mathrm{d}t \overset{\text{Hölder}}{\le} \int_0^1 \|v_t^k\|_{L^p(\mu_t^k)}\,\mathrm{d}t \overset{\text{Hölder}}{\le} \left(\int_0^1 \|v_t^k\|_{L^p(\mu_t^k)}^p\,\mathrm{d}t\right)^{\frac{1}{p}} \le C.$$

Since for $p > 1$ L^p is reflexive, E^k as a bounded sequence has a weakly converging subsequence FALSCHES ARGUMENT (eher Banach Alaoglu). Also, by a triangle inequality argument, we can show that μ_t^k converges uniformly in time to μ_t in W_p Explizit berechnen?. This gives us

$$\partial_t \mu_t + \nabla\cdot E = 0$$

in the distributional sense.
Action Functional noch definieren, Prop 5.18, eq 5.4 und Rest einfach noch hinschreiben. REST TODO

□

As the goal is to show that a solution of the static problem admits one to the dynamic problem and vice-versa, we also want to show a converse result. TODO Lemma 5.21 staten

Theorem 3.3.5. *If $(\mu_t)_{t\in[0,1]} \subset \mathscr{P}(\Omega)$ and for arbitrary t there exists $v_t \in L^p(\mu_t)$ satisfying $\int_0^1 \|v_t\|_{L^p(\mu_t)}dt < \infty$ and $\partial_t\mu_t + \nabla\cdot(v_t\mu_t) = 0$, then $|\mu'|(t) \le \|v_t\|_{L^p(\mu_t)}$ for a.e. t and $(\mu_t)_{t\in[0,1]}$ is absolutely continuous in $W_p(\Omega)$.*

Proof. Choosing a reparametrization as

$$s(t) := \int_0^t \|v_\tau\|_{L^p(\mu_\tau)} d\tau$$

for a given pair (μ, E) solving the continuity equation with $E_t = v_t \cdot \mu_t$ and $v_t \in L^p(\mu_t; \mathbb{R}^d)$, we get a uniform bound on $\|v_t\|_{L^p(\mu_t)}$ by and it also holds $\|E_t\| = \|v_t\|_{L^1(\mu_t)}$. Lemma ?? (5.21) guarantees the existence of a family $(\mu_t^\varepsilon)_t$ with a velocity field v_t^ε with the following properties:

1. μ^ε is Lipschitz in time and space, v^ε is Lipschitz in space and bounded, whilst being uniformly (bounded or continuous?) in time. $\mu^\varepsilon > 0$ everywhere.
2. We get μ_t^ε by following the flow of v^ε starting from μ_0^ε (see eq 5.5)
3. $\mu_t^\varepsilon \rightharpoonup \mu_t$ for all t and $E_t^\varepsilon \rightharpoonup E_t$ for a.e. t.
4. $\|v_t^\varepsilon\|_{L^p(\mu_t^\varepsilon)} \leq \mathscr{B}_p(\mu_t, E_t)$

DAS HIER INS LEMMA PACKEN UND HIER NUR STATEN
Taking a smooth function $\eta : \mathbb{R}^+ \to \mathbb{R}^+$ such that $\eta(t) = Ce^{-t}$ for $t \geq 1$ we construct $\eta^\varepsilon(z) = \varepsilon^{-d}\eta\left(\frac{|z|}{\varepsilon}\right)$, which lets us set

$$\mu_t^\varepsilon := \eta^\varepsilon * \mu_t, \quad E_t^\varepsilon := \eta^\varepsilon * E_t, \quad v_t^\varepsilon := \frac{E_t^\varepsilon}{\mu_t^\varepsilon}$$

With T_t being the flow generated by v^ε, we choose the transport plan between μ_t^ε and μ_{t+h}^ε as $\gamma = (T_t, T_{t+h})_\# \mu_0^\varepsilon \in \Gamma(\mu_t^\varepsilon, \mu_{t+h}^\varepsilon)$. For the p-th Wasserstein-Distance, we compute

$$\begin{aligned}
W_p^p(\mu_t^\varepsilon, \mu_{t+h}^\varepsilon) &\leq \int_{\Omega^2} |x-y|^p d\gamma(x,y) \\
&= \int_\Omega |T_t(x) - T_{t+h}(x)|^p d\mu_0^\varepsilon(x) \\
&\leq \int_\Omega \left(\int_t^{t+h} \left| \frac{d}{ds} T_s(x) \right| ds \right)^p d\mu_0^\varepsilon(x) \\
&\leq |h|^{\frac{p}{q}} \int_\Omega \int_t^{t+h} |v_s^\varepsilon(T_s(x))|^p ds d\mu_0^\varepsilon(x) \\
&= |h|^{\frac{p}{q}} \int_t^{t+h} \int_\Omega |v_s^\varepsilon(y)|^p d\mu_0^\varepsilon(y) ds \\
&\leq |h|^{\frac{p}{q}} \int_t^{t+h} \|v_s\|_{L^p(\mu_s)}^p ds.
\end{aligned}$$

The calculation

$$\begin{aligned}
|\mu'|(t) &= \lim_{h\to 0} \frac{W_p(\mu_{t+h}, \mu_t)}{|h|} = \lim_{h\to 0} \lim_{\varepsilon \to 0} \frac{W_p(\mu_{t+h}^\varepsilon, \mu_t^\varepsilon}{|h|} \\
&\leq \lim_{h\to 0} \left(\frac{1}{|h|} \int_t^{t+h} \|v_s\|_{L^p(\mu_t)}^p ds \right)^{\frac{1}{p}} = \|v_t\|_{L^p(\mu_t)}
\end{aligned}$$

yields the desired result and concludes the proof. □

Corollary 3.3.6. *The vector field v_t in Theorem ?? (das erste) satisfies $\|v_t\|_{L^p(\mu_t)} = |\mu'|(t)$.*

3.4 Numerical Analysis of the Saddle-Point Problem

Recall the definition of the Lagrangian

$$\mathcal{L}(\varphi, \rho, m) = \int_0^1 \int_\Omega \left(\frac{|m|^2}{2\rho} - \rho\varphi_t - m \cdot \nabla\varphi \right) \mathrm{d}x\mathrm{d}t + G(\varphi).$$

We want to solve the saddle-point problem

$$\inf_{\rho,m} \sup_{\varphi} \mathcal{L}(\varphi, \rho, m)$$

with φ being the designated Lagrange-multiplier.

Lemma 3.4.1. *We can rewrite $\frac{|m|^2}{2\rho}$ as the supremum of affine linear convex functions, namely*

$$\sup_{(a,b)\in K} (a\rho + b \cdot m), \quad K := \left\{ (a, b) \in \mathbb{R} \times \mathbb{R}^n \mid a + \frac{|b|^2}{2} \leq 0 \right\}$$

.

Proof. Let (a^*, b^*) obtain the supremum of the set K. This is possible since K is obviously non-empty. Since $\rho \geq 0 \in \mathscr{P}(\Omega)$, a^* needs to be as large as possible. Rewriting the constraint of the set, we have

$$a^* = \frac{|b|^2}{2} \implies \sup_{(a,b)\in K} (a\rho + b \cdot m) = \sup_{b\in\mathbb{R}^n} \left(\frac{-\rho|b|^2}{2} + b \cdot m \right)$$

Since the last expression is concave, we have the maximum by deriving and setting to zero and as a consequence

$$b^* = \frac{m}{\rho} \implies \sup_{(a,b)\in K} (a\rho + b \cdot m) = \frac{|m|^2}{2\rho}.$$

□

As a consequence we rewrite the optimization problem as a linear one, that is

$$\inf_{\rho,m} \sup_{\varphi,(a,b)\in K} \int_0^1 \int_\Omega [(a - \varphi_t)\rho + (b - \nabla\varphi) \cdot m] dx dt$$

Benamou-Brenier introduced an augmented Lagrangian (TODO wieso eigentlich). Define

$$\mu := \begin{pmatrix} \rho \\ m \end{pmatrix}, \quad q := \begin{pmatrix} a \\ b \end{pmatrix}$$

then we can rewrite

$$\inf_{\rho,m} \sup_{\varphi} \mathcal{L}(\varphi, \rho, m) = \sup_{\mu} \inf_{\varphi,q} F(q) + G(\varphi) + \langle \mu, \nabla_{t,x}\varphi - q \rangle$$

where $G(\varphi)$ is defined as above and $F(q) = \begin{cases} 0, & q \in K \\ \infty & \text{else} \end{cases}$

TODO Eq 42 nochmal den Beweis abschreiben

Algorithmus: Erste Update in ϕ, dann q, dann μ.
TODO zu ϕ – was ist das überhaupt für ein Objekt?
Updating the algoritm in φ is finding a solution for the expression

$$\arg\min_{\varphi} \mathcal{L}_r(\varphi, q^{n-1}, \mu) = \arg\min_{\varphi} G(\varphi) + \langle \mu^n, \nabla\varphi - q^{n-1}\rangle + \frac{r}{2}\|\nabla\varphi - q^{n-1}\|^2 =: J(\varphi)$$

To avoid confusion, we denote by ψ the test-direction for differentiation. For the optimality condition we compute

$$\frac{d}{d\varepsilon} J(\varphi^n + \varepsilon\psi)\Big|_{\varepsilon=0} = 0$$

and get for each term

$$\begin{aligned}
G(\varphi^n + \varepsilon\psi) &= \int_\Omega \left[(\nu(\varphi(1,x) + \varepsilon\psi(1,x)) - \mu(\varphi(0,x) + \varepsilon\psi(0,x))\right] dx \\
&= G(\varphi) + \varepsilon \int_\Omega (\nu\psi(1,x) - \mu\psi(0,x)) dx \\
\implies \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} G(\varphi^n + \varepsilon\psi) &= G(\psi) \\
\frac{d}{d\varepsilon}\langle \mu^n, \nabla(\varphi + \varepsilon\psi) - q^{n-1}\rangle &= \frac{d}{d\varepsilon}\langle \mu^n, \nabla\varphi^n + \varepsilon\psi - q^{n-1}\rangle \\
&= \frac{d}{d\varepsilon}\langle \mu^n, \nabla\varphi^n - q^{n-1}\rangle + \varepsilon\langle \mu^n, \nabla\psi\rangle \\
&= \langle \mu^n, \nabla\psi\rangle \\
\frac{d}{d\varepsilon}\frac{r}{2}\|\nabla(\varphi^n + \varepsilon\psi) - q^{n-1}\|^2 &= r\langle \nabla\varphi^n - q^{n-1}, \nabla\psi\rangle
\end{aligned}$$

To solve for φ^n we need to integrate by parts. To make notation easier we write

$$\int_\Omega (\nu\psi(1,x) - \mu\psi(0,x)) + \int (r(\nabla\varphi^n - q^{n-1}) + \mu^n) \cdot \nabla\psi = \int_\Omega (\nu\psi(1) - \mu\psi(0)) + \int A \cdot \nabla\psi = 0$$

with $A = \begin{pmatrix} A_t \\ A_x \end{pmatrix}$. Then

$$\begin{aligned}
\int A \cdot \nabla\psi &= -\int (\nabla \cdot A)\psi + \int_\Omega A_t(1)\psi(1) - \int_\Omega A_t(0)\psi(0) \\
\implies &-\int (\nabla \cdot A)\psi + \int_\Omega (\nu + A_t(1))\psi(1) - \int_\Omega (\mu + A_t(0))\psi(0) = 0
\end{aligned}$$

Since we assume periodic boundary conditions in space and Neumann boundary conditions in time, the boundary terms vanish and we calculate

$$\begin{aligned}
&\nabla \cdot (r(\nabla\varphi^n - q^{n-1}) + \mu^n) = 0 \\
&\iff r\Delta\varphi^n - r\nabla \cdot q^{n-1} + \nabla \cdot \mu^n = 0 \\
&\iff \Delta\varphi^n = \nabla \cdot q^{n-1} - \frac{1}{r}\nabla \cdot \mu^n
\end{aligned}$$

TODO Jetzt noch Lösungstheorie: Direkte Methode, Sobolev Raum, H1 Skalarprodukt
For practical purposes, Benamou-Brenier proposed a slightly perturbed version with a small $\varepsilon > 0$ given by

$$-r\Delta\varphi^n + r\varepsilon\varphi^n = \nabla \cdot (\mu^n - rq^{n-1})$$

Zu q:
Äquivalenz zeigen. q is given by solving

$$\inf_{(a,b)\in K} \{(a-\alpha^n)^2 + |b-\beta|^2 \mid a + \frac{|b|^2}{2} \leq 0\}$$

where $p^n(t,x) = \begin{pmatrix} \alpha^n(t,x) \\ \beta^n(t,x) \end{pmatrix} = \nabla\varphi^n(t,x) + \frac{\mu^n(t,x)}{r}$ The Lagrangian of this convex problem leads us directly to a analytical solution. Define

$$\mathcal{L}(a,b,\lambda) = (a-\alpha)^2 + (b-\beta)^2 + \lambda\left(a + \frac{|b|^2}{2}\right)$$

then the KKT conditions

$$\begin{cases} \partial_a L = 0 \\ \partial_b L = 0 \\ \lambda \geq 0 \\ a + \frac{|b|^2}{2} \leq 0 \\ \lambda\left(a + \frac{|b|^2}{2}\right) = 0 \end{cases}$$

are satisfied. Stationarity leads us to

$$\begin{cases} \partial_a L = 2(a-\alpha) + \lambda \overset{!}{=} 0 \implies a = \alpha - \frac{\lambda}{2} \\ \partial_b L = 2(b-\beta)\lambda\beta \overset{!}{=} 0 \implies b = \frac{2\beta}{2+\lambda} \end{cases}$$

In the case of $\lambda = 0$, then obviously $q^n = p^n$ (die zwei noch definieren).
For $\lambda > 0$ we need to solve

$$F(\lambda) := \alpha - \frac{\lambda}{2} + \frac{2|b|^2}{2\cdot(2+\lambda)^2} = 0$$

This problem can be solved iteratively using Newtons method

$$\lambda^{(k+1)} = \lambda^{(k)} - \frac{F(\lambda^{(k)})}{F'(\lambda^(k))}$$

with $F'(\lambda^{(k)}) = -\frac{1}{2} - \frac{4\beta^2}{(2+\lambda)^3}$. Note that in the case $\lambda > 0$ the solution is unique, since the problem is strictly convex.

4 Data Fitting

For our modeling purposes we modify the constraints and rewrite the initial Benamou-Brenier formulation as

$$\inf_{(v_t,\rho_t)} \int_0^1 \int_\Omega \frac{|v_t|^2}{2} \mathrm{d}\rho_t(x)\mathrm{d}t$$

$$\text{s.t.} \begin{cases} \partial_t \rho_t + \nabla \cdot \left(\rho_t v_t + \rho_t v_t^{wind}\right) = \alpha_{dif} \Delta \rho_t + f_t \quad \text{on } \Omega \times [0,1], \\ \left(\rho_t v_t + \rho_t v_t^{wind} - \alpha_{dif} \nabla \rho_t\right) \cdot n = 0 \quad \text{on } \partial\Omega \times [0,1], \\ \rho_{t=0} = \rho_0, \quad \rho_{t=1} = \rho_1. \end{cases}$$

If we recall

$$-\alpha_{dif} \Delta \rho_t = \nabla \cdot (-\alpha_{dif} \nabla \rho_t), \quad \nabla \rho_t = \rho_t \nabla \log \rho_t$$

we can reformulate the first constraint as

$$\begin{aligned} &\partial_t \rho_t + \nabla \cdot (\rho_t v_t + \rho_t v_t^{wind}) - \alpha_{dif} \Delta \rho_t = f_t \\ \iff &\partial_t \rho_t + \nabla \cdot (\underbrace{\rho_t (v_t - \alpha_{dif} \nabla \log \rho_t)}_{=: J_t} + \rho_t v_t^{wind}) = f_t. \end{aligned}$$

Using this notation and assuming $\rho_t > 0$ we can rewrite the above problem, using $v_t = \frac{J_t}{\rho_t} + \alpha_{dif} \nabla \log \rho_t$, as

$$\int_0^1 \int_\Omega \frac{|J_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t + \alpha_{dif} \int_0^1 \int_\Omega J_t \cdot \nabla \log \rho_t \,\mathrm{d}x\mathrm{d}t + \frac{\alpha_{dif}^2}{2} \int_0^1 \int_\Omega \underbrace{|\nabla \log \rho_t|^2 \rho_t}_{= \frac{|\nabla \rho_t|^2}{\rho_t}} \,\mathrm{d}x\mathrm{d}t.$$

Looking at the mixed term with the given constraints, we calculate by using partial integration and following the argumentation presented in $OSHER$

$$\begin{aligned} &\int_0^1 \int_\Omega J_t \cdot \nabla \log \rho_t \,\mathrm{d}x\mathrm{d}t \\ &= \int_0^1 \int_{\partial\Omega} \log \rho_t J_t \cdot n \,\mathrm{d}x\mathrm{d}t - \int_0^1 \int_\Omega \log \rho_t \nabla \cdot J_t \,\mathrm{d}x\mathrm{d}t \\ \overset{\text{substituting equations}}{=} &- \int_0^1 \int_{\partial\Omega} \log \rho_t \cdot \rho_t v_t^{wind} \cdot n \,\mathrm{d}x\mathrm{d}t - \int_0^1 \int_\Omega \log \rho_t f_t \,\mathrm{d}x\mathrm{d}t \\ &+ \underbrace{\int_0^1 \int_\Omega \log \rho_t \partial_t \rho_t}_{(*)} \,\mathrm{d}x\mathrm{d}t + \int_0^1 \int_\Omega \log \rho_t \nabla \cdot (\rho_t v_t^{wind}) \,\mathrm{d}x\mathrm{d}t \\ \overset{(*)}{=} &- \int_0^1 \int_{\partial\Omega} \log \rho_t \cdot \rho_t v_t^{wind} \cdot n \,\mathrm{d}x\mathrm{d}t - \int_0^1 \int_\Omega \log \rho_t f_t \,\mathrm{d}x\mathrm{d}t \\ &+ E(\rho_1) - E(\rho_0) - \int_0^1 \int_\Omega \nabla \log \rho_t \cdot \rho_t v_t^{wind} \,\mathrm{d}x\mathrm{d}t + \int_0^1 \int_{\partial\Omega} \log \rho_t \cdot \rho_t v_t^{wind} \cdot n \,\mathrm{d}x\mathrm{d}t. \end{aligned}$$

For $(*)$ we have used Fubini and integration by parts and obtain

$$\begin{aligned}
&\int_0^1 \int_\Omega \log \rho_t \frac{\partial \rho_t}{\partial t} \,\mathrm{d}x\mathrm{d}t = \int_\Omega \int_0^1 \log \rho_t \frac{\partial \rho_t}{\partial t} \,\mathrm{d}x\mathrm{d}t \\
=&\int_\Omega \rho_t \log \rho_t \Big|_{t=0}^{t=1} \mathrm{d}x - \int_0^1 \int_\Omega \rho_t \frac{\partial}{\partial t} \log \rho_t \,\mathrm{d}x\mathrm{d}t \\
=&\int_\Omega \left(\rho_1 \log \rho_1 - \rho_0 \log \rho_0\right) \,\mathrm{d}x - \underbrace{\int_0^1 \int_\Omega \frac{\partial \rho_t}{\partial t} \,\mathrm{d}x\mathrm{d}t}_{TODO} \\
=&\int_\Omega \rho_1(x) \log \rho_1(x) - \rho_0(x) \log \rho_0(x) \,\mathrm{d}x \\
=&E(\rho_1) - E(\rho_0) = \text{const}
\end{aligned}$$

thus giving the entire problem formulation as (TODO Frage an Juliane wegen gem Term/Nebenbed.)

$$\inf_{(\rho_t, J_t, \nabla \rho_t)} \int_0^1 \int_\Omega \frac{|J_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t + \alpha_{dif}^2 \int_0^1 \int_\Omega \frac{|\nabla \rho_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t$$

$$\text{s.t.} \begin{cases} \partial_t \rho_t + \nabla \cdot (\rho_t v_t + \rho_t v_t^{wind}) = f_t \quad \text{on } (0,1) \times \Omega \\ (\rho_t v_t + \rho_t v_t^{wind}) \cdot n = 0 \quad \text{on } (0,1) \times \partial\Omega \\ \rho(0,\cdot) = \rho_0, \quad \rho(1,\cdot) = \rho_1 \end{cases}$$

Adding two terms to obtain the desired model, one for allowing creation of mass and one for fitting data on a given curve, we finally receive

$$\begin{aligned}
&\inf_{(\rho_t, J_t, f_t, g_t)} \int_0^1 \int_\Omega \frac{|J_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t + \alpha_f^2 \int_0^1 \int_\Omega \frac{|f_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t \\
&+ \alpha_{dif}^2 \int_0^1 \int_\Omega \frac{|g_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t + \frac{\alpha_{inv}^2}{2} \int_0^1 \int_\Gamma |\rho_t - \rho_{data}|^2 \,\mathrm{d}x\mathrm{d}t
\end{aligned}$$

$$\text{s.t.} \begin{cases} \partial_t \rho_t + \nabla \cdot (J_t + \rho_t v_t^{wind}) = f_t \quad \text{on } \Omega \times (0,1) \\ (J_t + \rho_t v_t^{wind}) \cdot n = 0 \quad \text{on } \partial\Omega \times (0,1) \\ g_t = \nabla \rho_t \end{cases}$$

and we may impose the condition on the initial and final distribution as above. Using the boundary condition we get for the first constraint

$$\begin{aligned}
&\int_0^1 \int_\Omega \partial_t \rho_t \,\mathrm{d}x\mathrm{d}t + \int_0^1 \int_\Omega \nabla \cdot (J_t + \rho_t v_t^{wind}) \,\mathrm{d}x\mathrm{d}t - \int_0^1 \int_\Omega f_t \,\mathrm{d}x\mathrm{d}t \\
=&\int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} \left(\int_\Omega \rho_t \,\mathrm{d}x \right) \mathrm{d}t - \int_0^1 \int_\Omega f_t \,\mathrm{d}x\mathrm{d}t \\
=&\int_\Omega \rho_1 \,\mathrm{d}x - \int_\Omega \rho_0 \,\mathrm{d}x - \int_0^1 \int_\Omega f_t \,\mathrm{d}x\mathrm{d}t = 0.
\end{aligned}$$

Here, our goal is also to solve the saddle-point problem, for which we need to calculate the Lagrangian in the first place. For that, we define smooth Lagrange-multipliers $\phi_t : [0,1] \times \Omega \to \mathbb{R}$ and $\lambda_t : [0,1] \times \Omega \to \mathbb{R}^n$ and write

$$\begin{aligned}\mathcal{L}(\rho_t, J_t, f_t, g_t; \phi_t, \lambda_t) = &\int_0^1 \int_\Omega \frac{|J_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t + \alpha_f^2 \int_0^1 \int_\Omega \frac{|f_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t \\ &+ \alpha_{dif}^2 \int_0^1 \int_\Omega \frac{|g_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t + \frac{\alpha_{inv}^2}{2} \int_0^1 \int_\Gamma |\rho_t - \rho_{data}|^2 \,\mathrm{d}x\mathrm{d}t \\ &+ \int_0^1 \int_\Omega \phi_t(\partial_t \rho_t + \nabla \cdot (J_t + \rho_t v_t^{wind}) - f_t) \,\mathrm{d}x\mathrm{d}t \\ &+ \int_0^1 \int_\Omega \lambda_t \cdot (g - \nabla \rho_t) \,\mathrm{d}x\mathrm{d}t.\end{aligned}$$

Integrating by parts yields

$$\int_0^1 \int_\Omega \phi_t \partial_t \rho_t \,\mathrm{d}x\mathrm{d}t = \int_\Omega [\phi_1 \rho_1 - \phi_0 \rho_0] \,\mathrm{d}x - \int_0^1 \int_\Omega \rho_t \partial_t \phi_t \,\mathrm{d}x\mathrm{d}t.$$

For the divergence term we obtain using (Gaus divergence theorem?)

$$\begin{aligned}\int_0^1 \int_\Omega \phi_t \nabla \cdot (J + \rho_t v_t^{wind}) \,\mathrm{d}x\mathrm{d}t &= \int_0^1 \int_{\partial\Omega} \phi_t (J_t + \rho_t v_t^{wind}) \cdot n \,\mathrm{d}S\mathrm{d}t - \int_0^1 \int_\Omega \nabla \phi_t \cdot (J_t + \rho_t v_t^{wind}) \,\mathrm{d}x\mathrm{d}t \\ &\overset{\text{no flux boundary condition}}{=} - \int_0^1 \int_\Omega \nabla \phi_t (J_t + \rho_t v_t^{wind}) \,\mathrm{d}x\mathrm{d}t.\end{aligned}$$

and for the gradient constraint we get

$$\int_0^1 \int_\Omega \lambda_t \cdot (g_t - \nabla \rho_t) \,\mathrm{d}x\mathrm{d}t \overset{\text{divergence theorem}}{=} \int_0^1 \int_\Omega \lambda_t \cdot g_t \,\mathrm{d}x\mathrm{d}t + \int_0^1 \int_\Omega \rho_t \nabla \cdot \lambda \,\mathrm{d}x\mathrm{d}t - \int_0^1 \int_{\partial\Omega} \rho_t \lambda_t \cdot n \,\mathrm{d}S\mathrm{d}t.$$

To simplify calculations we assume $\lambda_t \cdot n = 0$ on $(0,1) \times \partial\Omega$. Putting those calculations together and introducing a real Lagrange multiplier κ_t, we end up with the final Lagrangian

$$\begin{aligned}\mathcal{L}(\rho_t, J_t, f_t, g_t, \phi_t, \lambda_t, \kappa_t) = &\int_0^1 \int_\Omega \frac{|J_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t + \alpha_f^2 \int_0^1 \int_\Omega \frac{|f_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t \\ &+ \alpha_{dif}^2 \int_0^1 \int_\Omega \frac{|g_t|^2}{2\rho_t} \,\mathrm{d}x\mathrm{d}t + \frac{\alpha_{inv}^2}{2} \int_0^1 \int_\Gamma |\rho_t - \rho_{data}|^2 \,\mathrm{d}x\mathrm{d}t \\ &- \int_0^1 \int_\Omega \rho_t \partial_t \phi_t + J_t \cdot \nabla \phi_t + \rho_t v_t^{wind} \cdot \nabla \phi_t + f_t \phi_t \,\mathrm{d}x\mathrm{d}t \\ &+ \underbrace{\int_\Omega [\phi_1 \rho_1 - \phi_0 \rho_0] \,\mathrm{d}x}_{G(\phi_t)} + \int_0^1 \int_\Omega \lambda_t \cdot g_t + \rho_t \nabla \cdot \lambda_t \,\mathrm{d}x\mathrm{d}t \\ &+ \kappa_t \left(\int_\Omega \rho_1 \,\mathrm{d}x - \int_\Omega \rho_0 \,\mathrm{d}x - \int_0^1 \int_\Omega f_t \,\mathrm{d}x\mathrm{d}t \right)\end{aligned}$$

and the saddle point problem then reads as

$$(\rho_t, J_t, f_t, g_t)(\phi_t, \lambda_t, \kappa_t)\mathcal{L}(\rho_t, J_t, f_t, g_t, \phi_t, \lambda_t, \kappa_t),$$

where the admissible set for (ρ_t, J_t, f_t, g_t) is given by

$$\mathcal{D}_{adm} = M^+_{[0,1]}(\Omega) \times M_{[0,1]}(\Omega, \mathbb{R}^d) \times M_{[0,1]}(\Omega) \times M_{[0,1]}(\Omega, \mathbb{R}^d).$$

To simplify notation, we introduce for $\kappa, \vartheta : [0,1] \times \Omega \to \mathbb{R}$

$$\langle \kappa, \vartheta \rangle = \int_0^1 \int_\Omega \kappa \vartheta \, \mathrm{d}x\mathrm{d}t, \quad |\kappa| = \sqrt{\langle \kappa, \kappa \rangle}.$$

For a primal-dual projection algorithm we introduce convex functions and their respective Legendre-Fenchel transform. We define the convex and lower semicontinuous functional

$$\Psi(u, v) := \begin{cases} \frac{|v|^2}{2u} & u > 0, \\ 0 & u = 0\,, v = 0 \\ +\infty & \text{else} \end{cases}$$

and the two convex and lower semicontinuous functions (as a sum of c and lsc functions)

$$F^{\alpha_f, \alpha_{dif}}(\rho_t, J_t, f_t, g_t) := \Psi(\rho_t, J_t) + \alpha_f^2 \Psi(\rho_t, f_t) + \alpha_{dif}^2 \Psi(\rho_t, g_t),$$
$$\overline{F}^{\alpha_f, \alpha_{dif}}(\rho_t, J_t, f_t, g_t) := F^{\alpha_f, \alpha_{dif}}(\rho_t, J_t, f_t, g_t) + \frac{\alpha_{inv}^2}{2} |\rho_t - \rho_{data}|^2,$$

where $F^{\alpha_f, \alpha_{dif}}$ is defined on $((0,1) \times \Omega) \setminus \Gamma$ and $\overline{F}^{\alpha_f, \alpha_{dif}}$ on Γ. We recall that for a convex function F can be rewritten as

$$F(x) = \sup_{x^*} \{ \langle x, x^* \rangle - F^*(x^*) \},$$

where F^* is the convex conjugate of F defined by

$$F^*(x^*) = \sup_{x} \{ \langle x, x^* \rangle - F(x) \}.$$

Impliclity we have used the fact, that $F^{**} = F$ for convex and lower semicontinuous F (Fenchel-Moreau). Using this principle, we have

$$F^{\alpha_f, \alpha_{dif}}(\rho_t, J_t, f_t, g_t) = \sup_{(\rho_t^*, J_t^*, f_t^*, g_t^*)} \{ \langle (\rho_t, J_t, f_t, g_t), (\rho_t^*, J_t^*, f_t^*, g_t^*) \rangle - (F^{\alpha_f, \alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*) \}$$

$$\overline{F}^{\alpha_f, \alpha_{dif}}(\rho_t, J_t, f_t, g_t) = \sup_{(\rho_t^*, J_t^*, f_t^*, g_t^*)} \{ \langle (\rho_t, J_t, f_t, g_t), (\rho_t^*, J_t*, f_t^*, g_t^*) \rangle - (\overline{F}^{\alpha_f, \alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*) \}.$$

Using this relation we rewrite the problem as (noch erwähnen, dass *-1 weil inf sup exchange)

$$\begin{aligned}(\rho_t, J_t, f_t, g_t)(\phi_t, \lambda_t, \kappa_t; \rho_t^*, J_t^*, f_t^*, g_t^*) & \int_0^1 \int_{((0,1)\times\Omega)\setminus\Gamma} (F^{\alpha_f, \alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*) \, \mathrm{d}x\mathrm{d}t \\ & + \int_\Gamma (\overline{F}^{\alpha_f, \alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*) \, \mathrm{d}x\mathrm{d}t - G(\phi_t) \\ & + \langle (\partial_t \phi_t + \nabla \phi_t \cdot v_t^{wind} - \rho_t^* - \nabla \cdot \lambda_t, \nabla \phi_t - J_t^*, \phi_t - f_t^*, -\lambda_t - g_t^*), (\rho_t, J_t, f \\ & - \kappa_t \left(\int_\Omega d\rho_1 - \int_\Omega d\rho_0 - \int_0^1 \int_\Omega f_t \, \mathrm{d}x\mathrm{d}t \right).\end{aligned}$$

We can evaluate the conjugate by the indicator function

$$\iota_C(x) := \begin{cases} 0 & x \in C, \\ +\infty & \text{else} \end{cases}$$

by

$$(F^{\alpha_f,\alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*) = \iota_{\left\{\rho_t^* + \frac{1}{2}\left(|J_t^*| + \frac{|f_t^*|}{\alpha_f^2} + \frac{|g_t^*|}{\alpha_{dif}^2}\right) \leq 0\right\}}.$$

The given set is calculated similarly to what we did for the classical Benamou-Brenier formulation meaning that for fixed ρ_t, we maximize

$$F^*(\rho_t^*, J_t^*, f_t^*, g_t^*) = \sup_{(\rho_t, J_t, f_t, g_t)} \left\{ \rho_t \rho_t^* + J_t \cdot J_t^* + f_t f_t^* + g_t g_t^* - \frac{|J_t|^2}{2\rho_t} - \alpha_f^2 \frac{|f_t|^2}{2\rho_t} - \alpha_{dif}^2 \frac{|g_t|^2}{2\rho_t} \right\}$$

seperately in every variable. This is done by setting the derivative with respect to each primal variable to zero, since for each variable the set is concave. This leads to the component-wise computation

$$\begin{cases} \frac{\partial}{\partial J_t} J_t \cdot J_t^* - \frac{|J_t|^2}{2\rho_t} \stackrel{!}{=} 0 \implies J_t = J_t^* \rho_t & \implies \sup_{J_t} \left\{ J_t \cdot J_t^* - \frac{|J_t|^2}{2\rho_t} \right\} = \frac{\rho_t |J_t^*|^2}{2} \\ \frac{\partial}{\partial f_t} f_t \cdot f_t^* - \alpha_f^2 \frac{|f_t|^2}{2\rho_t} \stackrel{!}{=} 0 \implies f_t = \frac{\rho_t}{\alpha_f^2} f_t^* & \implies \sup_{f_t} \left\{ f_t \cdot f_t^* - \alpha_f^2 \frac{|f_t|^2}{2\rho_t} \right\} = \frac{\rho_t |f_t^*|^2}{2\alpha_f^2} \\ \frac{\partial}{\partial g_t} g_t \cdot g_t^* - \alpha_{dif}^2 \frac{|g_t|^2}{2\rho_t} \stackrel{!}{=} 0 \implies g_t = \frac{\rho_t}{\alpha_{dif}^2} g_t^* & \implies \sup_{g_t} \left\{ g_t \cdot g_t^* - \alpha_{dif}^2 \frac{|g_t|^2}{2\rho_t} \right\} = \frac{\rho_t |g_t|^2}{2\alpha_{dif}^2} \end{cases}$$

To calculate

$$(\overline{F}^{\alpha_f,\alpha_{dif}})^* = \sup_{\rho_t, J_t, f_t, g_t} \left\{ \rho_t \rho_t^* + J_t \cdot J_t^* + f_t f_t^* + g_t \cdot g_t^* - \frac{|J_t|^2}{2\rho_t} - \alpha_f^2 \frac{|f_t|}{2\rho_t} - \alpha_{dif}^2 \frac{|g_t|^2}{2\rho_t} - \frac{\alpha_{inv}^2}{2} |\rho_t - \rho_{data}|^2 \right\}$$

we again evaluate the first derivative in each primal variable and the first optimality condition with respect to ρ_t gives

$$\begin{aligned} &\rho_t^* + \frac{1}{2}\left(|J_t^*|^2 + \frac{|f_t^*|^2}{\alpha_f^2} + \frac{|g_t^*|^2}{\alpha_{dif}^2} \right) - \alpha_{inv}^2 (\rho_t - \rho_{data}) = 0 \\ \iff &\rho_t = \rho_{data} + \frac{1}{2\alpha_{inv}^2} \left(2\rho_t^* + |J_t^*|^2 + \frac{|f_t^*|^2}{\alpha_f^2} + \frac{|g_t^*|^2}{\alpha_{dif}^2} \right). \end{aligned}$$

This allows us to rewrite

$$\begin{aligned} (\overline{F}^{\alpha_f,\alpha_{dif},\alpha_{inv}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*) &= \sup_{\rho_t} \rho_t \left(\rho_t^* + \frac{|J_t^*|^2}{2} + \frac{|f_t^*|^2}{2\alpha_f^2} + \frac{|g_t^*|^2}{2\alpha_{dif}^2} \right) - \frac{\alpha_{inv}^2}{2} |\rho_t - \rho_{data}|^2 \\ &= \rho_{data} \left(\rho_t^* + \frac{|J_t^*|^2}{2} + \frac{|f_t^*|^2}{2\alpha_f^2} + \frac{|g_t^*|^2}{2\alpha_{dif}^2} \right) + \frac{1}{2\alpha_{inv}^2} \left(\rho_t^* + \frac{|J_t^*|^2}{2} + \frac{|f_t^*|^2}{2\alpha_f^2} + \frac{|g_t^*|^2}{2\alpha_{dif}^2} \right)^2. \end{aligned}$$

For a regulization parameter $r_1 > 0$ we obtain the Augmented Lagrangian defined as

$$\begin{aligned}
&\mathcal{L}_{aug}(\rho_t, J_t, f_t, g_t, \phi_t, \lambda_t, \rho_t^*, J_t^*, f_t^*, g_t^*) := \int_{((0,1)\times\Omega)\setminus\Gamma} (F^{\alpha_f,\alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*)\,\mathrm{d}x\mathrm{d}t \\
&+ \int_\Gamma (\overline{F}^{\alpha_f,\alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*)\,\mathrm{d}x\mathrm{d}t - G(\phi_t) \\
&+ \langle(\partial_t\phi_t + \nabla\phi_t \cdot v_t^{wind} - \rho_t^* - \nabla\cdot\lambda_t, \nabla\phi_t - J_t^*, \phi_t - f_t^*, -\lambda_t - g_t^*), (\rho_t, J_t, f_t, g_t)\rangle \\
&+ \frac{r_1}{2}|\partial_t\phi_t + \nabla\phi_t \cdot v_t^{wind} - \rho_t^* - \nabla\cdot\lambda_t|^2 + \frac{r_1}{2}(|\nabla\phi_t - J_t^*|^2 + |\phi_t - f_t^*|^2 + |-\lambda_t - g_t^*|^2) \\
&+ \kappa_t\left(\int_\Omega \rho_1\,\mathrm{d}x - \int_\Omega \rho\,\mathrm{d}x - \int_0^1\int_\Omega f_t\,\mathrm{d}x\mathrm{d}t\right)
\end{aligned}$$

To solve this problem numerically, the Benamou Brenier Algorithm can be rewritten to fit the modified problem. As in the classical ALG2 alrogithm, we update each tuple of variables independently. For a shorter notation we define

$$\begin{aligned}
\eta_{\rho^*} &:= \partial_t\phi_{m+1} + \nabla\phi_{m+1}\cdot v_t^{wind} - \nabla\cdot\lambda_{m+1} + \frac{1}{r_1}\rho_m \\
\eta_{J^*} &:= \nabla\phi_{m+1} + \frac{1}{r_1}J_m \\
\eta_{f^*} &:= \phi_{m+1} + \frac{1}{r_1}f_m \\
\eta_{g^*} &:= -\lambda_{m+1} + \frac{g_m}{r_1} \\
\eta_\Omega &:= (\eta_{\rho^*}, \eta_{J^*}, \eta_{f^*}, \eta_{g^*}) \\
z_m^* &:= (\rho_m^*, J_m^*, f_m^*, g_m^*)
\end{aligned}$$

Step 1: Update (ϕ_t)
We want to solve

$$\inf_{(\phi_{m+1})} \mathcal{L}(\rho_t, J_t, f_t, g_t, \phi_t, \lambda_t, \kappa_t, \rho_t^*, J_t^*, f_t*, g_t^*)$$

by calculating the first variation with respect to ϕ and setting it equal to zero. In particular we aim to evaluate the pertubation for all $\psi \in H^1((0,1)\times\Omega)$

$$\frac{d}{d\varepsilon}\mathcal{L}_{aug}(\phi_t + \varepsilon\psi_t)\Big|_{\varepsilon=0} = 0.$$

For a more concise representation we define

$$\phi_\varepsilon = \phi_{m+1} + \varepsilon\phi$$

and look at each summand individually. Starting with the endpoint term we obtain

$$\begin{aligned}
G(\phi_\varepsilon) &= \langle\phi_{m+1}(1) - \varepsilon\phi_{m+1}(0), \rho_1\rangle + \langle\phi_{m+1}(1) + \varepsilon\phi_{m+1}(0), \rho_0\rangle \\
\implies \frac{d}{d\varepsilon}G(\phi_\varepsilon)\Big|_{\varepsilon=0} &= \langle\psi(1), \rho_1\rangle - \langle\psi(0), \rho_0\rangle = G(\psi).
\end{aligned}$$

Similarly we calculate for the scalar products in the augmented Lagrangian

$$\begin{cases} \frac{d}{d\varepsilon}\langle \partial_t\phi_\varepsilon + \nabla\phi_\varepsilon \cdot v_t^{wind} - \rho_m^* - \nabla\cdot\lambda_m, \rho_m\rangle\Big|_{\varepsilon=0} & = \langle \partial_t\psi + \nabla\psi\cdot v_t^{wind}, \rho_m\rangle, \\ \frac{d}{d\varepsilon}\langle \nabla\phi_\varepsilon - J_m^*, J_m\rangle\Big|_{\varepsilon=0} & = \langle \nabla\psi, J_m\rangle, \\ \frac{d}{d\varepsilon}\langle \phi_\varepsilon - f_m^*, f_m\rangle\Big|_{\varepsilon=0} & = \langle \psi, f_m\rangle. \end{cases}$$

For the quadratic terms we use the identity

$$\frac{d}{d\varepsilon}\frac{1}{2}|a+\varepsilon b|^2\Big|_{\varepsilon=0} = \langle a, b\rangle.$$

Then the first variations of the quadratic terms yield

$$\begin{cases} \frac{d}{d\varepsilon}\frac{r_1}{2}\left|\partial_t\phi_\varepsilon + \nabla\varepsilon\cdot v_t^{wind} - \rho_m^* - \nabla\cdot\lambda_m\right|^2\Big|_{\varepsilon=0} \\ = r_1\langle \partial_t\phi_{m+1} + \nabla\phi_{m+1}\cdot v_t^{wind} - \rho_m^* - \nabla\lambda_m, \partial_t\psi + \nabla\phi\cdot v_t^{wind}\rangle \\ \frac{d}{d\varepsilon}\frac{r_1}{2}|\nabla\phi_\varepsilon - J_m^*|^2\Big|_{\varepsilon=0} = r_1\langle \nabla\phi_{m+1} - J_m^*, \nabla\psi\rangle \\ \frac{d}{d\varepsilon}\frac{r_1}{2}|\phi_\varepsilon - f_m^*|^2\Big|_{\varepsilon=0} = r_1\langle \phi_{m+1} - f_m^*, \psi\rangle \end{cases}$$

Rearranging the individual scalar products then enables us to write out the the complete first variation of the augmented Lagrangian, which we want to set to zero, i.e.

$$\begin{aligned} 0 = &-\langle\psi(1),\rho_1\rangle + \langle\psi(0),\rho_0\rangle + \langle \partial_t\psi + \nabla\psi\cdot v_t^{wind}, \rho_m\rangle + \langle\nabla\psi, J_m\rangle + \langle\psi, f_m\rangle \\ &+ r_1\Big(\langle \partial_t\phi_{m+1}, \partial_t\psi + \nabla\psi\cdot v_t^{wind}\rangle \\ &+ \langle \nabla\phi_{m+1}\cdot v_t^{wind}, \partial_t\psi + \nabla\psi\cdot v_t^{wind}\rangle - \langle \rho_m^*\partial_t\psi - \nabla\psi\cdot v_t^{wind}\rangle \\ &- \langle \nabla\cdot\lambda_m, \partial_t\psi + \nabla\psi\cdot v_t^{wind}\rangle + \langle\nabla\phi_{m+1}, \nabla\phi\rangle \\ &- \langle J_m^*, \nabla\psi\rangle + \langle\phi_{m+1},\psi\rangle - \langle f_m^*,\psi\rangle\Big). \end{aligned}$$

Grouping terms containing ϕ_{m+1} lets us write this expression as a bilinear and linear form respectively. For that, define both

$$\begin{aligned} A(\phi_{m+1},\psi_t) :=& r_1\Big(\langle\nabla\phi_{m+1},\nabla\psi_t\rangle + \langle\phi_{m+1},\psi_t\rangle + \langle\nabla\phi_{m+1}\cdot v_t^{wind}, \partial_t\psi_t + \nabla\psi_t\cdot v_t^{wind}\rangle \\ &+ \langle\partial_t\phi_{m+1}, \partial_t\psi_t + \nabla\psi_t\cdot v_t^{wind}\rangle\Big) \\ F_m(\psi_t) :=& G(\psi) + \langle \partial_t\psi_t + \nabla\psi_t\cdot v_t^{wind}, r_1\rho_m^* + r_1\nabla\cdot\lambda_m - \rho_m\rangle \\ &+ \langle\nabla\psi_t, r_1J_m^* - J_m\rangle + \langle\psi_t, r_1f_m^* - f_m\rangle. \end{aligned}$$

Hence the update can be compactly described as finding ϕ_{m+1} such that

$$A(\phi_{m+1},\psi_t) = F_m(\psi_t) \quad \forall\psi\in H^1((0,1)\times\Omega).$$

Step 1b: Update (λ_t)

$$\lambda_{m+1} = \lambda_m - \left(\int_\Omega d\rho_1 - \int_\Omega d\rho_0 - \int_0^1\int_\Omega f_m\,\mathrm{d}x\mathrm{d}t\right)$$

TODO κ tauschen – λ an welcher Stelle optimieren?

Step 2: Update $(\rho_t^*, J_t^*, f_t^*, \nabla\rho_t^*)$

The new iterates are defined as the solution of

$$\begin{aligned}\inf_{(\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*, g_{m+1}^*)} &\int_{((0,1)\times\Omega)\setminus\Gamma} (F^{\alpha_f,\alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*)\,\mathrm{d}x\mathrm{d}t + \int_\Gamma (\overline{F}^{\alpha_f,\alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, g_t^*)\,\mathrm{d}x\mathrm{d}t \\ &- \langle \rho_t^*, \rho_m + r_1(\partial_t\phi_{m+1} + \nabla\phi_{m+1}\cdot v_t^{wind} - \nabla\cdot\lambda_{m+1}\rangle + \frac{r_1}{2}\|\rho_t^*\|^2 \\ &- \langle J_t^*, J_m + r_1\nabla\phi_{m+1}\rangle + \frac{r_1}{2}|J_t^*|^2 - \langle f_t^*, f_m + r_1\phi_{m+1}\rangle + \frac{r_1}{2}|f_t^*|^2 \\ &- \langle g_t^*, g_m - r_1\lambda_{m+1}\rangle + \frac{r_1}{2}|g_t^*|^2\end{aligned}$$

where terms constant in $(\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*, g_{m+1}^*)$ are neglected. We optimize this problem in $((0,1)\times\Omega)\setminus\Gamma$ and in Γ independently. Taking the variables defined above, the minimization in the former reads as

$$\begin{aligned}\inf_{(\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*, g_{m+1}^*)} \iota_{\left\{\rho_t^* + \frac{1}{2}\left(|J_t^*| + \frac{|f_t^*|}{\alpha_f} + \frac{|g_t|}{\alpha_{dif}}^2\right)\le 0\right\}} &- r_1\langle \eta_\Omega, (\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*, g_{m+1}^*)\rangle \\ &+ \frac{r_1}{2}(|\rho_{m+1}^*|^2 + |J_{m+1}^*|^2 + |f_{m+1}^*|^2 + |g_{m+1}^*|^2)\end{aligned} \tag{3}$$

with the optimal solution η_Ω, if it is admissible. If not, we introduce another Lagrange multiplier $\mu > 0$ and since the problem has a constant term with respect to η_Ω we can rewrite as

$$\inf_{(\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*, g_{m+1}^*)} \frac{1}{2}|z_{m+1}^* - \eta_\Omega|_\Omega^2$$

The new Lagrangian for this problem is

$$\begin{aligned}\mathcal{L}(\rho_t^*, J_t^*, f_t^*, g_t^*, \mu) = &\frac{1}{2}(|\rho^* - \eta_{\rho^*}|^2 + |J^* - \eta_{J^*}|^2 + |f^* - \eta_{f^*}|^2 + |g^* - \eta_{g^*}|^2) \\ &+ \mu\rho^* + \frac{\mu}{2}\left(|J^*|^2 + \frac{|f^*|^2}{\alpha_f} + \frac{|g^*|}{\alpha_{dif}}\right)\end{aligned}$$

Define

$$x := |J^*|, \quad y := |f^*|, \quad u := |g^*|$$

and rewrite the active constraint as

$$\rho^* = -\frac{1}{2}\left(x^2 + \frac{y^2}{\alpha_f} + \frac{u^2}{\alpha_{dif}}\right).$$

Substituting this expression into

$$\min_{x,y,u}\frac{1}{2}\left[\left(-\frac{1}{2}\left(x^2 + \frac{y^2}{\alpha_f} + \frac{u^2}{\alpha_{dif}}\right) - \eta_{\rho^*}\right)^2 + (x - |\eta_{J^*}|)^2 + (y - |\eta_{f^*}|)^2 + (u - |\eta_{g^*}|)^2\right]$$

and taking the derivatives with respect to x, y and u yields the equations

$$\begin{aligned}0 &= x^3 + \frac{1}{\alpha_f}xy^2 + \frac{1}{\alpha_{dif}}xu^2 + 2(1 + \eta_{\rho^*})x - 2|\eta_{J^*}| \\ 0 &= y^3 + \alpha_f x^2 y + \frac{\alpha_f}{\alpha_{dif}}yu^2 + 2(\alpha_f^2 + \alpha_f\eta_{\rho^*})y - 2\alpha_f^2|\eta_{f^*}| \\ 0 &= u^3 + \alpha_{dif}x^2u + \frac{\alpha_{dif}}{\alpha_f}y^2u + 2\alpha_{dif}(\alpha_{dif} + \eta_{\rho^*})u - 2\alpha_{dif}^2|\eta_{g^*}|.\end{aligned}$$

Solving these equations for x, y, u we recover

$$\begin{cases} \rho^* = -\frac{1}{2}\left(x^2 + \frac{y^2}{\alpha_f} + \frac{u^2}{\alpha_{dif}}\right), \\ J^* = x\frac{\eta_{J^*}}{|\eta_{J^*}|}, \\ f^* = y\frac{\eta_{f^*}}{|\eta_{f^*}|}, \\ g^* = u\frac{\eta_{g^*}}{|\eta_{g^*}|}, \end{cases}$$

whenever the denominators are non-zero for an analytical solution. This can numerically be solved also by using Newton's method. On Γ we can solve the system that is obtained by calculating the gradient of (**??**)

$$\begin{pmatrix} \rho_{data} + 2\left(\frac{1}{\alpha_{inv}} - \frac{\alpha_{inv}}{2}\right)\left(\rho^* + \frac{1}{2}|J^*|^2 + \frac{1}{\alpha_f}|f^*|^2 + \frac{1}{2\alpha_{dif}}|g^*|^2\right) + r_1(\rho^* - \eta_{\rho^*}) \\ \rho_{data}J^* + 2\left(\frac{1}{\alpha_{inv}} - \frac{\alpha_{inv}}{2}\right)\left(\rho^* + \frac{1}{2}|J^*|^2 + \frac{1}{\alpha_f}|f^*|^2 + \frac{1}{2\alpha_{dif}}|g^*|\right)J^* + r_1(J^* - \eta_{J^*}) \\ \frac{1}{\alpha_f}\rho_{data}f^* + \frac{1}{\alpha_f}\left(\frac{1}{\alpha_{inv}} - \frac{\alpha_{inv}}{2}\right)\left(\rho^* + \frac{1}{2}|J^*|^2 + \frac{1}{\alpha_f}|f^*|^2 + \frac{1}{2\alpha_{dif}}|g^*|\right)f^* + r_1(f^* - \eta_{f^*}) \\ \frac{1}{\alpha_{dif}}\rho_{data}g^* + \frac{1}{\alpha_{dif}}\left(\frac{1}{\alpha_{inv}} - \frac{\alpha_{inv}}{2}\right)\left(\rho^* + \frac{1}{2}|J^*|^2 + \frac{1}{\alpha_f}|f^*|^2 + \frac{1}{2\alpha_{dif}}|g^*|^2\right)g^* + r_1(g^* - \eta_{g^*}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Step 3: Update (ρ_t, J_t, f_t, g_t)
As in step 2, we neglegt terms constant to the respective variables and need to solve

$$\sup_{(\rho_t, J_t, f_t, g_t)} \langle (\partial_t\phi_t + \nabla\phi_t \cdot v_t^{wind} - \rho_t^* - \nabla\cdot\lambda_t, \nabla\phi_t - J_t^*, \phi_t - f_t^*, -\lambda_t - g_t^*), (\rho_t, J_t, f_t, g_t)\rangle$$

which is done by a single gradient descent step in each variable with step size r_1, i.e.

$$\begin{aligned} \rho_{m+1} &:= \rho_m + r_1(\partial_t\phi_{m+1} + \nabla\phi_{m+1}\cdot v_t^{wind} - \rho_{m+1}^* - \nabla\cdot\lambda_{m+1}) \\ J_{m+1} &:= J_m + r_1(\nabla\phi_{m+1} - J_{m+1}^*) \\ f_{m+1} &:= f_m + r_1(\phi_{m+1} - f_{m+1}^*) \\ g_{m+1} &:= g_m + r_1(-\lambda_{m+1} - g_{m+1}^*). \end{aligned}$$

Augmented Lagrangian Method with diffusion term $m < it_max^{ALG}$ and $err^{ALG} > tol$
Solve $A(\phi_{m+1}) = F^m$ for ϕ_{m+1}
Update λ_{m+1}
$\eta_{\rho^*} := \partial_t\phi_{m+1} + \nabla\phi_{m+1}\cdot v_t^{wind} + \frac{1}{r_1}\rho_m$
$\eta_{J^*} := \nabla\phi_{m+1} + \frac{1}{r_1}J_m$
$\eta_{f^*} := \phi_{m+1} + \frac{1}{r_1}f_m$
$\eta_{\nabla\phi^*} := \frac{1}{r_1}\nabla\rho_m$
$\eta_\Omega := (\eta_{\rho^*}, \eta_{J^*}, \eta_{f^*}, \eta_{\nabla\rho^*})$
$(t,x) \in (0,1)\times\Omega\setminus\Gamma\ \eta_{\rho^*} + \frac{1}{2}\left(|\eta_{J^*}|^2 + \frac{|\eta_{f^*}|^2}{\alpha_f} + \frac{|\eta_{\nabla\rho^*}|^2}{\alpha_{dif}}\right) \le 0$
$(\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*, \nabla\rho_{m+1}^*) = (\eta_{\rho^*}, \eta_{J^*}, \eta_{f^*}, \eta_{\nabla\rho^*})$
Solve nonlinear system for $x, y, u \ge 0$
$\rho_{m+1}^* \leftarrow -\frac{1}{2}\left(x^2 + \frac{y^2}{\alpha_f} + \frac{u^2}{\alpha_{dif}}\right)$
$J_{m+1}^* \leftarrow x\frac{\eta_{J^*}}{|\eta_{J^*}|}$
$f_{m+1}^* \leftarrow y\frac{\eta_{f^*}}{|\eta_{f^*}|}$

$\nabla\rho^*_{m+1} \leftarrow u\frac{\eta_{\nabla\rho^*}}{|\eta_{\nabla\rho^*}|}$ $(t,x) \in \Gamma$
Solve $\nabla_z \left[(F^{\alpha_f,\alpha_{dif},\alpha_{inv}} + \frac{r_1}{2})^*|z^* - \eta_\Omega|^2\right] = 0$ for $z^* = (\rho^*, J^*, f^*, \nabla\rho^*)$
Set $z^*_{m+1} \leftarrow z^*$
Update $(\rho_{m+1}, J_{m+1}, f_{m+1}, \nabla\rho_{m+1})$ by
$\rho_{m+1} \leftarrow \rho_m + r_1\left(\partial_t\phi_{m+1} + v_t^{wind}\cdot\nabla\phi_{m+1} - \rho^*_{m+1}\right)$
$J_{m+1} \leftarrow J_m + r_1\left(\nabla\phi_{m+1} - J^*_{m+1}\right)$
$f_{m+1} \leftarrow f_m + r_1\left(\phi_{m+1} - f^*_{m+1}\right) + \lambda_{m+1}$
$\nabla\rho_{m+1} \leftarrow \nabla\rho_m - r_1\nabla\rho^*_{m+1}$
Compute while-loop conditions
$err^{ALG} \leftarrow \max\left\{\left|\partial_t\phi_{m+1} + v_t^{wind}\cdot\nabla\phi_{m+1} - \rho^*_{m+1}\right|_\infty, \left|\nabla\phi_{m+1} - J^*_{m+1}\right|_\infty, \left|\phi_{m+1} - f^*_{m+1}\right|_\infty, \left|\nabla\rho^*_{m+1}\right|_\infty\right\}$
$m \leftarrow m+1$

5 Numerische Ergebnisse

Erklärung von FEM, das Vorgehen der Lösung, Code erklären.